



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.AT.6C

Write and Evaluate Numerical Expressions with Exponents

Lesson 6-4 • Teacher Copy

Name: Type your name

Date: Today's date

Class: Class period

My Notes

Level 2 Standard



TODAY'S OBJECTIVES

Content Objective: I can evaluate algebraic expressions by substituting values for the variables.

Language Objective: I can explain my steps using the words variable, expression, evaluate, and substitute.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Evaluate Expressions

Variable

Expression

Evaluate

Substitute

Coefficient

Like terms



Tap any word to see what it means and a picture.

1

— Find the value of an expression by substituting numbers and following the order of operations.

2

A letter that stands for an unknown number, like x , is a

3

A group of numbers, variables, and operations with no equal sign is an

4

To find the value of an expression for given values is to

it.

5

To replace a variable with a number is to

.

- 6 A number multiplied by a variable, like the 4 in $4y$, is a

.

- 7 Terms with the same variable part, like $2x$ and $6x$, are

.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How much will the ride cost? How does the expression model this situation?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Evaluate Expressions** — Find the value of an expression by substituting numbers and following the order of operations.

Evaluar expresiones — Hallar el valor de una expresión sustituyendo números y siguiendo el orden de las operaciones.

- ☐ **Variable** — A letter that stands for a number that is unknown or can change.

Variable — Una letra que representa un número desconocido o que puede cambiar.

- ☐ **Expression** — A math phrase with numbers and letters, but no equal sign.

Expresión — Una frase matemática con números y letras, pero sin signo igual.

- ☐ **Evaluate** — To find the value by putting numbers in for the letters.

Evaluar — Encontrar el valor poniendo números en lugar de las letras.

- ☐ **Substitute** — To put a number in place of a letter.

Sustituir — Poner un número en lugar de una letra.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The ride costs \$ ____ because $2.50 + 1.75(\text{ }) = \text{ }$. The 2.50 is the ____ and 1.75 is the ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Substitute the number of speakers for s , then evaluate.

Evidence: Students substitute $s = 5$ ($150 + 12 \times 5 = 210$), explain 150 is the fixed venue fee and 12 is the cost per speaker, and note s can be any number.

Reasoning: This evidence connects the definition of expression to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The ride costs \$ ____ because $2.50 + 1.75(\text{____}) = \text{____}$. The 2.50 is the ____ and 1.75 is the ____.
- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
- I know because ____.

Mi afirmación es ____.

Lo sé porque ____.

- So ____.

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Evaluate Expressions
2. **Notes 2:** Variable
3. **Notes 3:** Expression
4. **Notes 4:** Evaluate
5. **Notes 5:** Substitute
6. **Notes 6:** Coefficient
7. **Notes 7:** Like terms
8. **Watch for this mistake:** *A common mistake in Evaluate Expressions is treating a coefficient like an addend instead of a factor — for example, evaluating $3x + 7$ when $x = 4$ as $3 + 4 + 7 = 14$ instead of the correct $3(4) + 7 = 12 + 7 = 19$. A number written right next to a variable always means multiply, never add. Before you submit, ask: "Did I multiply the coefficient by the value I substituted, or did I accidentally add it?"*
9. **Try It 1:** A. 36 — x^2 means x times itself. Substitute $x = 6$: $6^2 = 6 \times 6 = 36$.
10. **Try It 2:** — *Two x -tiles stand for $2x$ and three unit tiles stand for 3. When $x = 4$, each x -tile is worth 4, so the two x -tiles are worth 8 and the three unit tiles add 3 more, giving 11.*